7 Conclusion

We obtain the surprising result that model edit success is essentially unrelated to where factual information is stored in models, as measured by Causal Tracing. Faced with this result, we attempt to reconnect tracing-based localization with edit success by introducing four variants of the Error Injection problem using the CounterFact dataset. We find that edit success and tracing effects correlate best in our Fact Forcing setting. However, even in this case, tracing effects explain only a small fraction of the variance in editing performance, while the choice of edit layer is a much more important factor. This suggests that, counterintuitively, better mechanistic understanding of how pretrained language models work may not always translate to insights about how to best change their behavior.

8 Limitations

We note a few limitations of the experiments conducted in this paper:

(1) We work only with the CounterFact and ZSRE datasets, which we use as short English prompts with factual completions corresponding to a specific set of relations between subject and object entities. This is a basic form of factual knowledge, and localization and editing analysis may yield different trends for other forms of knowledge.

(2) We work with two autoregressive Transformers chosen for their representativeness of large language models that show a capacity for expressing factual knowledge in response to natural language prompts. However, the conclusions from our analysis may not generalize to models larger than GPT-J (6B parameters) that are known to exhibit phase changes in their behavior under prompting.

(3) We use a particular set of localization and editing methods, including representation denoising and zeroing at the layer level and layer-level MLP editing methods that inject new facts or amplify or erase existing facts. Our conclusions may not necessarily hold for the breadth of localization and editing methods from work related to this paper, and one should be cautious in applying our conclusions beyond our experimental setting.

9 Broader Impacts

It is possible that increased mechanistic understanding of models improves our ability to edit them at some point in the future. In fact, we consider it unlikely that interpretability results never give insight into improving model editing methods. Thus, to the extent that model editing is a dual use methodology, which could be used to inject harmful beliefs or dangerous knowledge into models, interpretability results may enhance the effectiveness of these malicious use cases. However, these concerns are relatively far removed from our analysis, which focuses on the connection between localization and editing performance. Ultimately, we hope that studies of mechanistic interpretability and model editing improve our ability to control language models.

Acknowledgements

We thank Kevin Meng and David Bau for helpful discussions of the methods used in this paper and the conclusions we reach from our experiments. Additionally, we thank Jasmijn Bastings and Lucas Dixon for feedback on writing and presentation of results, as well as the paper's reviewers for their useful comments. This work was conducted while Peter Hase was a student researcher at Google.

References

[1] David Bau, Steven Liu, Tongzhou Wang, Jun-Yan Zhu, and Antonio Torralba. Rewriting a deep generative model. In *European conference on computer vision*, pages 351–369. Springer, 2020. URL https://arxiv.org/pdf/2007.15646.pdf.

[2] David Bau, Jun-Yan Zhu, Hendrik Strobelt, Agata Lapedriza, Bolei Zhou, and Antonio Torralba. Understanding the role of individual units in a deep neural network. *Proceedings of the National*

Academy of Sciences, 117(48):30071–30078, 2020. URL https://www.pnas.org/doi/pdf/10.1073/pnas.1907375117.

[3] Tolga Bolukbasi, Adam Pearce, Ann Yuan, Andy Coenen, Emily Reif, Fernanda Viégas, and Martin Wattenberg. An interpretability illusion for bert. *arXiv preprint arXiv:2104.07143*, 2021. URL https://arxiv.org/pdf/2104.07143.pdf.

[4] Stephen Casper, Shlomi Hod, Daniel Filan, Cody Wild, Andrew Critch, and Stuart Russell. Graphical clusterability and local specialization in deep neural networks. In *ICLR 2022 Workshop on PAIR*, 2022. URL https://arxiv.org/pdf/2110.08058v2.pdf.

[5] Pattarawat Chormai, Jan Herrmann, Klaus-Robert Müller, and Grégoire Montavon. Disentangled explanations of neural network predictions by finding relevant subspaces. *arXiv preprint arXiv:2212.14855*, 2022. URL https://arxiv.org/pdf/2212.14855.pdf.

[6] Róbert Csordás, Sjoerd van Steenkiste, and Jürgen Schmidhuber. Are neural nets modular? inspecting functional modularity through differentiable weight masks. *arXiv preprint arXiv:2010.02066*, 2020. URL https://arxiv.org/pdf/2010.02066.pdf.

[7] Audrey Cui, Ali Jahanian, Agata Lapedriza, Antonio Torralba, Shahin Mahdizadehaghdam, Rohit Kumar, and David Bau. Local relighting of real scenes. *arXiv preprint arXiv:2207.02774*, 2022. URL https://arxiv.org/pdf/2207.02774.pdf.

[8] Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, and Furu Wei. Knowledge neurons in pretrained transformers. In *ACL*, 2022. URL https://arxiv.org/pdf/2104.08696.pdf.

[9] Nicola De Cao, Wilker Aziz, and Ivan Titov. Editing factual knowledge in language models. In *EMNLP*, pages 6491–6506. Association for Computational Linguistics, November 2021. URL https://aclanthology.org/2021.emnlp-main.522.

[10] Nicola De Cao, Leon Schmid, Dieuwke Hupkes, and Ivan Titov. Sparse interventions in language models with differentiable masking. In *EMNLP BlackboxNLP Workshop*, 2021. URL https://arxiv.org/pdf/2112.06837.pdf.

[11] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. https://transformer-circuits.pub/2021/framework/index.html.

[12] Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In *EMNLP*, 2021. URL https://arxiv.org/pdf/2012.14913.pdf.

[13] Mor Geva, Avi Caciularu, Kevin Ro Wang, and Yoav Goldberg. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. *arXiv preprint arXiv:2203.14680*, 2022. URL https://arxiv.org/pdf/2203.14680.pdf.

[14] Amirata Ghorbani, James Wexler, James Y Zou, and Been Kim. Towards automatic concept-based explanations. *Advances in Neural Information Processing Systems*, 32, 2019. URL https://arxiv.org/pdf/1902.03129.pdf.

[15] Peter Hase, Mona Diab, Asli Celikyilmaz, Xian Li, Zornitsa Kozareva, Veselin Stoyanov, Mohit Bansal, and Srinivasan Iyer. Do language models have beliefs? methods for detecting, updating, and visualizing model beliefs. *arXiv preprint arXiv:2111.13654*, 2021. URL https://arxiv.org/pdf/2111.13654.pdf.

[16] Evan Hernandez, Sarah Schwettmann, David Bau, Teona Bagashvili, Antonio Torralba, and Jacob Andreas. Natural language descriptions of deep visual features. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/pdf?id=NudBMY-tzDr.

[17] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory sayres. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV). In Jennifer Dy and Andreas Krause, editors, *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 2668–2677. PMLR, 10–15 Jul 2018. URL https://proceedings.mlr.press/v80/kim18d.html.

[18] Yair Lakretz, German Kruszewski, Theo Desbordes, Dieuwke Hupkes, Stanislas Dehaene, and Marco Baroni. The emergence of number and syntax units in lstm language models. In *NAACL-HLT*, 2019. URL https://arxiv.org/pdf/1903.07435.pdf.

[19] Yair Lakretz, Dieuwke Hupkes, Alessandra Vergallito, Marco Marelli, Marco Baroni, and Stanislas Dehaene. Mechanisms for handling nested dependencies in neural-network language models and humans. *Cognition*, 213:104699, 04 2021. doi: 10.1016/j.cognition.2021.104699. URL https://arxiv.org/ftp/arxiv/papers/2006/2006.11098.pdf.

[20] Omer Levy, Minjoon Seo, Eunsol Choi, and Luke Zettlemoyer. Zero-shot relation extraction via reading comprehension. In *Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL 2017)*, pages 333–342, Vancouver, Canada, August 2017. Association for Computational Linguistics. doi: 10.18653/v1/K17-1034. URL https://aclanthology.org/K17-1034.

[21] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual knowledge in gpt. In *NeurIPS 2022*, 2022. URL https://arxiv.org/pdf/2202.05262.pdf.

[22] Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. Mass-editing memory in a transformer. *arXiv preprint arXiv:2210.07229*, 2022. URL https://arxiv.org/pdf/2210.07229.pdf.

[23] Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. Efficient estimation of word representations in vector space. In *arXiv preprint arXiv:1301.3781*, 2013. URL https://arxiv.org/pdf/1301.3781.pdf.

[24] Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D Manning. Fast model editing at scale. *arXiv preprint arXiv:2110.11309*, 2021. URL https://arxiv.org/pdf/2110.11309.pdf.

[25] Eric Mitchell, Charles Lin, Antoine Bosselut, Christopher D Manning, and Chelsea Finn. Memory-based model editing at scale. In *International Conference on Machine Learning*, pages 15817–15831. PMLR, 2022. URL https://arxiv.org/pdf/2206.06520.pdf.

[26] Jesse Mu and Jacob Andreas. Compositional explanations of neurons. *Advances in Neural Information Processing Systems*, 33:17153–17163, 2020. URL https://proceedings.neurips.cc/paper/2020/file/c74956ffb38ba48ed6ce977af6727275-Paper.pdf.

[27] Chris Olah, Arvind Satyanarayan, Ian Johnson, Shan Carter, Ludwig Schubert, Katherine Ye, and Alexander Mordvintsev. The building blocks of interpretability. *Distill*, 2018. doi: 10.23915/distill.00010. https://distill.pub/2018/building-blocks.

[28] Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. Language models as knowledge bases? In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2463–2473, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1250. URL https://aclanthology.org/D19-1250.

[29] Alec Radford, Rafal Jozefowicz, and Ilya Sutskever. Learning to generate reviews and discovering sentiment. *arXiv preprint arXiv:1704.01444*, 2017. URL https://arxiv.org/pdf/1704.01444.pdf.